

Evaluating Spectral Interpretability in Brazilian Portuguese Audio Deepfakes via Causal Occlusion

Como as pistas acústicas de um detector XLS-R mudam entre zero-shot e adaptação leve — TODO: confirmar/traduzir

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ABSTRACT

State-of-the-art audio deepfake detectors are self-supervised black boxes trained overwhelmingly on English anti-spoofing data, leaving open which acoustic cues they actually exploit when applied to other languages such as Brazilian Portuguese. We present a population-level acoustic explainability study of a wav2vec 2.0 / XLS-R detector on the BRSpeech-DF corpus, contrasting the zero-shot model (D_{zs}) with a lightly adapted variant (D_{ad} : frozen XLS-R plus a logistic read-out). We characterise the detectors along two complementary axes over the same analysis set: an *associative* axis (intra-class Spearman correlation between 16 log-mel band-energy descriptors and the spoof score) and a *causal* axis (spectral occlusion on the *same* mel-band grid, with per-band bootstrap confidence intervals), plus a confirmatory layer (Welch and Levene tests under Benjamini–Hochberg control). Because both axes share one frequency grid, we can directly quantify their agreement. Light adaptation lowers the EER from 25.8% to 20.9% and reshapes the cues: the causal spectral footprint changes from a diffuse, broad-band pattern (occluding almost any band lowers the spoof score) to a localised low-frequency reliance with an emergent mid-band cue that favours bona fide, while the associative profile weakens and moves toward mid-band energy. For the zero-shot detector the causally influential bands and the most associated ones broadly coincide (band-wise Spearman $\rho = 0.69$), a correspondence that adaptation largely removes ($\rho = 0.10$). Error-linked acoustic differences are statistically significant after FDR correction for both detectors. All results are reproducible from fixed seeds and released code.

KEYWORDS

audio deepfake detection, anti-spoofing, explainability, Spearman correlation, spectral occlusion, log-mel band energy, cross-lingual, wav2vec 2.0, XLS-R, Brazilian Portuguese

1 INTRODUCTION

Synthetic and manipulated speech has become cheap to produce and hard to detect by ear, motivating automatic anti-spoofing detectors. The strongest of these are built on self-supervised learning (SSL) front-ends such as wav2vec 2.0 / XLS-R, which achieve remarkable accuracy but behave as black boxes. Crucially, they are trained overwhelmingly on English-centric benchmarks (e.g. ASVspoof), so when deployed on other languages they may rely on cues that are entangled with the phonetics of the training language rather than with genuine artefacts of synthesis. For Brazilian Portuguese (pt-BR) in particular, little is known about *which acoustic properties* drive

a detector’s decision, or whether those properties are language-biased.

We therefore ask: *which acoustic cues does an SSL deepfake detector rely on to decide, and how do those cues change before versus after a light adaptation to Brazilian Portuguese?* To answer this without retraining the representation, we compare two detectors that share the same frozen XLS-R front-end — a zero-shot detector (D_{zs}) using the checkpoint’s original head, and a lightly adapted detector (D_{ad}) that retrains only a logistic read-out on pt-BR embeddings. This design isolates the effect of adaptation on the *use* of existing representations, providing a controlled “before/after” contrast for acoustic explanation.

Our contributions are: (i) a reproducible protocol for *population-level* acoustic explainability of black-box deepfake detectors in pt-BR; (ii) two complementary lines of evidence — *associative* (intra-class Spearman correlation over log-mel band-energy descriptors) and *causal* (spectral occlusion with bootstrap uncertainty) — computed on the *same* analysis set and, crucially, on a *shared* mel-frequency grid so their agreement is directly measurable; (iii) a measurement of how these cues shift under light adaptation; and (iv) fully reproducible tables and figures with fixed seeds and released code.

We frame the study around one research question: *which acoustic cues does the detector rely on, and how do these cues shift after a light adaptation to Brazilian Portuguese?* We evaluate two detectors sharing a frozen XLS-R front-end: a zero-shot detector (D_{zs}) and a lightly adapted one (D_{ad}).

One condition must hold for the comparison to be meaningful: adaptation must improve detection. We require $EER(D_{ad}) < EER(D_{zs})$ before any cue analysis; otherwise the “before/after” contrast is reported as *insufficient adaptation*. Because our associative analysis is a direct, model-agnostic rank correlation (rather than attributions from a fitted surrogate), it carries no separate surrogate-fidelity precondition.

We investigate three hypotheses:

- **H1 (associative cue shift).** The acoustic-association profile over log-mel band-energy features differs between D_{zs} and D_{ad} . We measure association as the *intra-class Spearman rank correlation* between each band-energy descriptor and $P(\text{spoof})$ (FDR-controlled per detector), and compare the profiles *descriptively* across detectors; we make no significance claim on the between-detector contrast.
- **H2 (causal spectral shift).** The frequency bands whose occlusion reduces $P(\text{spoof})$ differ between D_{zs} and D_{ad} . Because occlusion is an intervention on the input, we report

the resulting causal profiles *with quantified uncertainty* (95% bootstrap confidence intervals on the per-band mean drop): a band whose interval excludes zero exerts a reliable causal effect on the score. We stop short of a formal significance test across detectors.

- **H3 (error-linked acoustic differences).** For the band-energy features most salient by intra-class Spearman $|\rho|$, distributions differ across decision quadrants: in mean between true-negative and false-positive clips (Welch’s *t*-test) and in variance between true-positive and false-negative clips (Levene’s test). H3 is tested *inferentially*, controlling the false discovery rate (Benjamini–Hochberg, $q < 0.05$).

Because H1 and H2 are expressed on the *same* mel-band grid, we additionally quantify their *agreement* for each detector as the band-wise Spearman correlation between the causal occlusion profile and the associative $|\rho|$ profile: a high value means the bands the detector causally relies on are also the ones most associated with its score.

2 RELATED WORK

2.1 Self-Supervised Front-Ends for Anti-Spoofing

Tak et al. [4] established the now-dominant recipe for spoofing and deepfake detection: a wav2vec 2.0 self-supervised front-end, fine-tuned with aggressive data augmentation, feeding a lightweight back-end classifier. This family of models — and its multilingual XLS-R variants — underpins most state-of-the-art anti-spoofing systems and is the class of detector we study. Their strength, however, comes with opacity: the learned representation is a high-dimensional black box, and the training data are predominantly English (ASVspoof-style). Our work takes such a detector as given and asks what acoustic information its decisions actually track when transferred to Brazilian Portuguese.

2.2 Explainability in Audio Deepfake Detection

A growing line of work opens up these detectors. Grinberg et al. [3] probe what a detector attends to in the *time* domain, showing that decisions can hinge on localised temporal regions rather than on globally distributed evidence. Fullwood and Monrose [2] interpret how detector behaviour *changes* under data augmentation, linking training-time interventions to shifts in what the model relies on. Our study is complementary along three axes. First, it is *spectral* rather than temporal: we localise cues in frequency via causal occlusion. Second, it is *population-level*: instead of explaining individual clips, we characterise cue usage across the whole analysis set with statistical uncertainty. Third, it targets a *cross-lingual* setting (pt-BR) and measures how cues move under adaptation. This empirical, interpretability-focused study is also complementary to, and non-overlapping with, our group’s prior PRISMA systematic review.

2.3 Cross-Lingual Generalization and Real-World Deployment

Channing et al. [1] argue that robust real-world deepfake detection requires closing an “explainability gap”: detectors that generalise

across domains and languages must be understood, not just benchmarked. Cross-lingual transfer is precisely where language-biased cues become a liability, since a detector tuned to English phonetics may latch onto spurious correlates when scored on another language. This motivates our focus on pt-BR and on *how* the acoustic cues of a fixed representation are re-weighted when the read-out is adapted to the target language.

3 METHODOLOGY

3.1 Detectors Under Study

Both detectors share the same frozen wav2vec 2.0 / XLS-R front-end (nii-yamagishilab/mms-300m-anti-deepfake); nothing inside the SSL encoder is retrained, so any behavioural difference is attributable to the read-out alone. D_{zs} (zero-shot) applies the checkpoint’s original anti-spoofing head directly to pt-BR. D_{ad} (lightly adapted) discards that head and instead mean-pools the frozen encoder outputs into a single 1024-dimensional embedding per clip, on top of which we train a logistic-regression classifier that emits a calibrated $P(\text{spoof})$. Both detectors thus map one clip to one score, enabling an identical downstream analysis. The “before/after” contrast is only licensed if adaptation actually helps: we verify $EER(D_{ad}) < EER(D_{zs})$ before conducting any cue analysis (Section 5.1).

3.2 Pre-processing

Each clip is converted to mono, resampled to 16 kHz, cropped or padded to a fixed window of 64,600 samples (≈ 4.04 s), and normalised following Tak et al. [4]. A key methodological point is that we extract acoustic features from *exactly* the pre-processed signal that enters the detector, not from the raw waveform: the input normalisation neutralises absolute level, so explaining the model on raw audio would misattribute cues (in particular for the log-energy coefficient). Analysing the signal the model actually “hears” keeps the explanation faithful.

3.3 Acoustic Features (log-mel band energy)

We summarise each clip with the *log-mel band energy* on the same eight mel-spaced bands used by the causal occlusion (Section 3.5), spanning 20–7900 Hz. On 25 ms windows with a 10 ms hop we integrate the STFT power within each band, take the logarithm, and collapse the temporal axis into two statistics per band — the mean (μ) and the standard deviation (σ) — yielding $8 \times 2 = 16$ features. The means capture how much energy sits in each part of the spectrum, while the standard deviations capture temporal *variability* within the clip, which turns out to be discriminative for the error analysis. This choice is deliberate: unlike cepstral coefficients, band energies are localised in frequency, so the associative analysis operates on the *same* axis as the causal occlusion analysis and the two can be compared band by band.

3.4 Associative Analysis (Spearman)

The associative analysis measures, for each detector, how each band-energy descriptor is associated with the continuous score $P(\text{spoof})$. We use Spearman’s rank correlation ρ , computed *within each class*

(bona fide and spoof separately). The intra-class restriction is essential: because the score separates the two classes, pooling them would manufacture a large spurious correlation for almost any feature that merely differs between classes. Within a class, ρ instead reflects genuine monotonic covariation between the band energy and the detector's confidence. We control the false discovery rate across features per detector (Benjamini–Hochberg) and compare the resulting $|\rho|$ profiles between D_{zs} and D_{ad} *descriptively*. This estimator is model-agnostic and transparent — it needs no fitted surrogate and therefore has no fidelity precondition — at the cost of being univariate (it does not model feature interactions).

3.5 Causal Analysis (Spectral Occlusion)

The causal analysis intervenes directly on the input. We partition the spectrum into eight mel-spaced bands spanning 20–7900 Hz and, one band at a time, remove it with a band-stop filter applied to the waveform (operating on the waveform avoids the lossy mel→audio inversion). The filtered clip is passed through the *same* detector, and we record the drop in $P(\text{spoof})$ relative to the intact-audio baseline; a positive drop means the band was *sustaining* the spoof score. We apply this to a quadrant-stratified subset (600 clips per band) and quantify uncertainty with a per-band bootstrap (95% CI on the mean drop): a band whose interval excludes zero exerts a reliable causal effect. Because occlusion is an intervention on the audio, the occlusion analysis yields interventional evidence that converges with — but is methodologically independent of — the associative analysis. Crucially, both analyses are defined on the *same* eight bands, so this agreement can be measured band by band rather than merely asserted (see Section 3.6). For implementation safety, audio is resampled to 16 kHz before filtering, keeping all band edges below the 8 kHz Nyquist limit.

3.6 Convergence of the Two Analyses

Because the associative and causal analyses share one mel-band grid, we summarise their agreement per detector with a single band-wise Spearman correlation between the causal profile (the magnitude of the mean occlusion drop per band) and the associative profile (the per-band maximum $|\rho|$ over statistics and classes). With only eight bands this correlation is a descriptive summary rather than a powered test; a high value indicates that the bands the detector causally relies on are also those most associated with its score.

3.7 Confirmatory Statistics

To avoid a fishing expedition over all 16 loosely motivated t -tests, the confirmatory layer restricts formal testing to the few band-energy features already flagged as most salient by the associative analysis (largest intra-class $|\rho|$). On those features we run two contrasts tied to the detector's error structure: a Welch t -test for a *mean* difference between correctly and incorrectly rejected clips (TN vs. FP), and a Levene test for a *variance* difference between caught and missed spoofs (TP vs. FN). All p -values are corrected for multiple comparisons with the Benjamini–Hochberg procedure, and we report significance as $q < 0.05$. This is the only part of the study framed inferentially; the associative and causal analyses remain descriptive.

Table 1: Experimental configuration.

Parameter	Value
Sample rate	16,000 Hz
Fixed window	64,600 samples (≈ 4.04 s)
Front-end	wav2vec 2.0 / XLS-R (frozen)
Checkpoint	nii-yamagishilab/mms-300m-anti-deepfake
Adaptation head (D_{ad})	logistic regression on 1024-d embeddings
Association measure	intra-class Spearman ρ (band energy vs. $P(\text{spoof})$)
Acoustic features	8 mel bands $\times$ (mean, std) = 16
Frequency bands	8 mel-spaced, 20–7900 Hz (shared with occlusion)
Dataset	AKCIT-Deepfake/BRSpeech-DF
Corpus scale	459,137 clips (76,644 bona fide / 382,493 spoof)
Train per class (D_{ad} head)	1,500 (balanced)
Analysis set (test)	1,320 bona fide + 1,500 spoof
Positive class	spoof
Seed	42

Table 2: Decision-quadrant sizes on the analysis set at the EER threshold, per detector (TP/TN: correct spoof/bona-fide; FP/FN: errors).

Detector	TP	TN	FP	FN
D_{zs} (zero-shot)	1,113	979	341	387
D_{ad} (adapted)	1,186	1,044	276	314

4 EXPERIMENTAL SETUP

Dataset and sampling rationale. BRSpeech-DF is a large-scale Brazilian-Portuguese anti-spoofing corpus (459,137 clips: 76,644 bona fide and 382,493 spoofed, the latter carrying a generator-method attribute), split into train/validation/test. Its test partition alone holds 1,320 bona fide and 6,587 spoofed clips. Because our objective is population-level acoustic *explainability* rather than leaderboard-style benchmarking, we deliberately analyse a compact, class-controlled subset instead of the full corpus. We take the *entire* bona fide test split (1,320 clips — the scarce class) matched with 1,500 spoofed test clips, and fit the D_{ad} logistic head on an independent, class-balanced training subset (1,500 per class); test clips are never seen during adaptation, precluding leakage. This scale is a design choice, not a data limitation. First, every inferential claim (H3: Welch and Levene under Benjamini–Hochberg control) and every interval estimate (H2: per-band bootstrap CIs) is already well powered at this N : at the EER operating point the rare error quadrants (FP, FN) still retain a few hundred clips each, enough to resolve small-to-moderate effects ($d \approx 0.25$ – 0.3 at 80% power), so additional clips would add negligible statistical resolution. Second, exhausting the scarce bona fide class while keeping a comparable spoof sample keeps the four decision quadrants (TP/TN/FP/FN) and the EER threshold meaningful, which the corpus's native $\approx 1:5$ bona fide/spoof imbalance would otherwise distort. Third, the per-clip analysis (band-stop spectral occlusion with a per-band bootstrap) scales with N , so a focused subset preserves full reproducibility at tractable cost. Scaling the analysis to the full corpus and stratifying it by spoof generator is left to future work.

Table 3: Master table schema (one row per audio clip; $_zs$ = zero-shot, $_ad$ = adapted).

Column	Description
sample_id	unique clip identifier
ground_truth	0 = bonafide, 1 = spoof
spoof_algorithm	generator/attack (null if absent)
p_spoof_{zs,ad}	continuous $P(\text{spoof})$ per detector
pred_{zs,ad}	decision at EER threshold
quadrant_{zs,ad}	TP / TN / FP / FN
band{k}_mean	mean log-energy of k -th band ($k = 1..8$)
band{k}_std	std of log-energy of k -th band ($k = 1..8$)

Table 4: Detection performance on the analysis set (performance_table).

Detector	EER	MCC	Accuracy
D_{zs} (zero-shot)	25.8%	0.483	0.742
D_{ad} (adapted)	20.9%	0.581	0.791

The four decision quadrants at the EER threshold are reported in Table 2: adaptation increases both correct quadrants (TP, TN) and shrinks both error quadrants (FP, FN), yet each rare quadrant still retains a few hundred clips, supporting the confirmatory tests of H3. All experiments run in a reproducible Docker environment on a single NVIDIA A100 GPU, with fixed seeds.

5 RESULTS

5.1 Zero-Shot vs. Adapted Performance

On the analysis set of 2,820 clips (1,500 spoof and 1,320 bona fide), the light adaptation lowers the EER from 25.8% to 20.9% — a 4.9-point absolute (about 19% relative) reduction — while MCC rises from 0.483 to 0.581 and accuracy from 0.742 to 0.791 (Table 4). Consistently, the D_{ad} DET curve in Figure 1 lies inside that of D_{zs} across operating points, so the EER pre-condition ($\text{EER}(D_{ad}) < \text{EER}(D_{zs})$) holds and the before/after cue comparison is warranted. Adaptation also reduces both error types: false positives fall from 341 to 276 and false negatives from 387 to 314. Because only the linear read-out is retrained on top of the frozen XLS-R front-end, these gains reflect a re-weighting of pre-existing representations rather than new features learned for Brazilian Portuguese.

6 EXPLAINABILITY ANALYSIS

6.1 Acoustic Association via Spearman

Figure 2 compares the per-band association profiles of D_{zs} and D_{ad} . For the zero-shot detector, the score is most strongly associated with *low-band energy means*: the 20–282 Hz band (intra-class $\rho = +0.45$ in spoof, $+0.39$ in bona fide) and the 282–639 Hz band ($\rho = +0.41$ in both classes), with secondary mid-band contributions at 1.79–2.69 and 2.69–3.93 kHz ($\rho \approx +0.38$ in spoof). All are positive: more band energy co-varies with a higher spoof score. Under adaptation the profile both *weakens* and *moves upward in frequency*: D_{ad} 's strongest

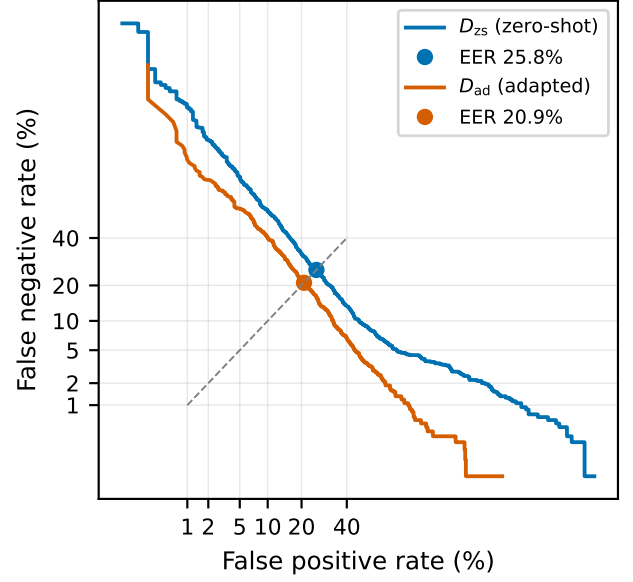


Figure 1: Detection error trade-off (DET) curves for D_{zs} and D_{ad} on the analysis set; the marker on each curve is the operating point at the Equal Error Rate. A curve closer to the origin indicates better detection.

associations are mid-band energy means that turn *negative* in bona-fide clips (1.79–2.69 kHz $\rho = -0.34$; 2.69–3.93 kHz $\rho = -0.33$; 1.13–1.79 kHz $\rho = -0.29$), together with mid-band energy *variability* in spoof clips (2.69–3.93 kHz $\sigma \rho = +0.25$). Descriptively, then, D_{zs} is associated mainly with low-band energy, whereas D_{ad} 's (weaker) univariate associations sit in the mid band. All associations cited above are FDR-significant per detector ($q \ll 0.01$); we make no significance claim on the between-detector contrast itself.

6.2 Spectral Occlusion Profiles

Figures 3 and 4 report the causal spectral profiles (each band occluded on 600 clips; 95% bootstrap CI). For D_{zs} the effect is broad-band and one-signed: occluding almost any band lowers $P(\text{spoof})$ — seven of eight bands have a CI excluding zero (all but 639–1125 Hz) — so energy across the whole spectrum contributes to the spoof decision. The effect is largest at low frequencies (20–639 Hz, mean drop ≈ 0.04) and at 1788–2693 Hz (0.037). Adaptation turns this diffuse pattern into a band-selective one and reverses one mid band: for D_{ad} the positive contribution concentrates at 20–282 Hz (0.060) and 639–1125 Hz (0.052), the 282–639 Hz band becomes null (CI includes zero), and 1125–1788 Hz *changes sign* (mean drop = -0.016 , CI $[-0.025, -0.007]$): occluding it *raises* $P(\text{spoof})$, so that band instead contributes to a bona-fide decision. In causal terms, adaptation concentrates the detector's spectral reliance on low frequencies and introduces a mid-band cue that favours bona fide, absent in the zero-shot model.

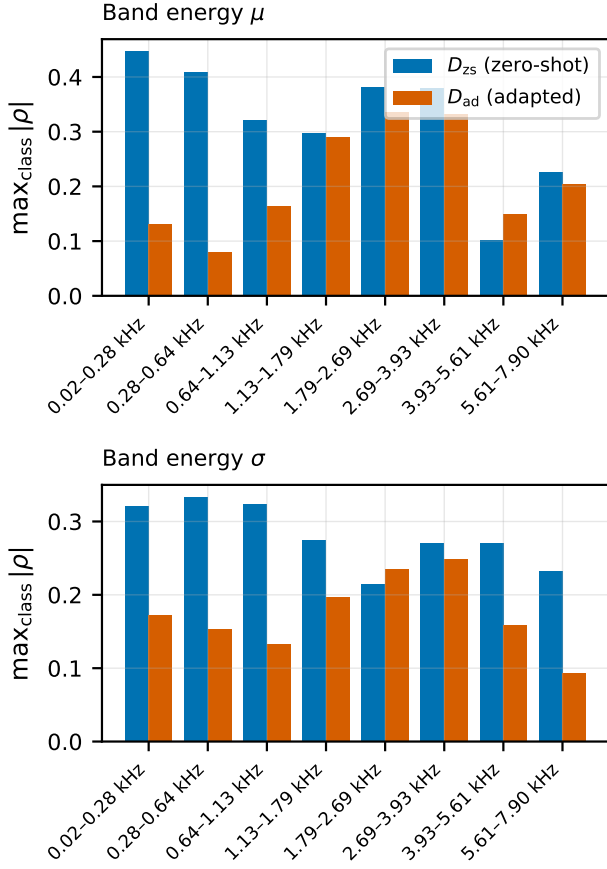


Figure 2: Acoustic-association profile of the log-mel band-energy features, measured as the intra-class Spearman $|\rho|$ between each band descriptor and $P(\text{spoof})$ (maximum over the bona fide/spoof classes), split into mean (μ , top) and standard-deviation (σ , bottom) descriptors and ordered by frequency band along the x -axis. We read the between-detector shift descriptively (H1).

6.3 Convergence of the Two Analyses

Because both analyses live on the same eight bands, we can ask whether the bands the detector *causally* relies on are the same ones its score is *associated* with (Figure 5). For the zero-shot detector the two profiles are broadly consistent: the band-wise Spearman correlation is $\rho = 0.69$, with both the causal drops and the associations largest in the low bands. With only eight bands this is suggestive rather than conclusive ($p = 0.06$, not significant at the 0.05 level), so we read it as descriptive support rather than proof. For the adapted detector the correspondence essentially vanishes ($\rho = 0.10$, $p = 0.82$): its causal footprint (low bands plus a mid-band cue that favours bona fide) is no longer tracked by the univariate association ranking. We do not speculate on the source of this divergence; we simply note that, for D_{ad} , band-level claims should rest on the causal analysis rather than the associative one.

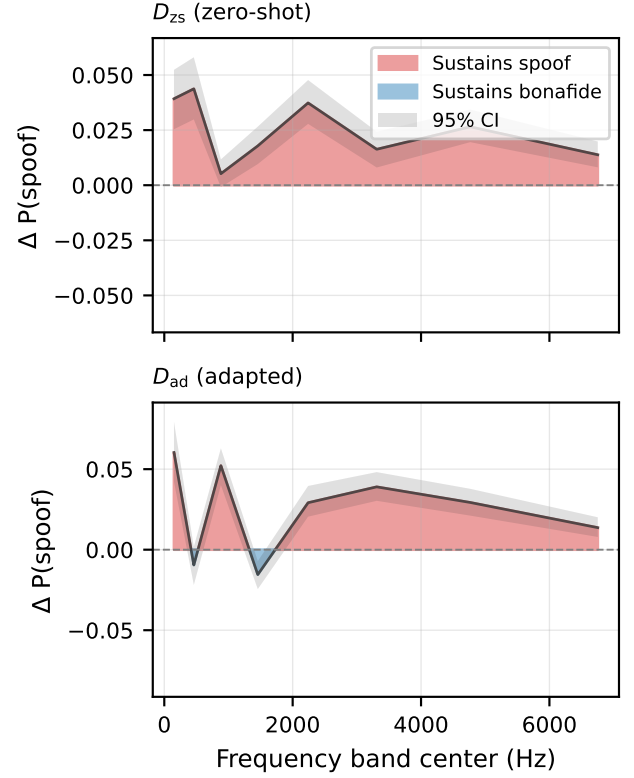


Figure 3: Divergent spectral occlusion profiles (D_{zs} top, D_{ad} bottom). Red area: occluding the band lowers $P(\text{spoof})$, so the band contributes to the spoof decision; blue: occlusion raises it, so the band contributes to a bona-fide decision; grey ribbon: 95% bootstrap CI of the per-band mean drop. Bands whose CI excludes zero have a reliable causal effect (H2).

6.4 Confirmatory Tests

Figure 6 shows the quadrant-wise distributions with FDR-corrected significance. Testing the ten most $|\rho|$ -salient band-energy features across decision quadrants, 9 of 20 tests are significant after Benjamini-Hochberg correction ($q < 0.05$) for D_{zs} and 13 of 20 for D_{ad} . The mean contrast between correct and incorrect rejections (TN vs. FP, Welch) is largest for the low-band energy means in D_{zs} – 282–639 Hz ($t = -16.1$, $q = 2 \times 10^{-49}$) and 20–282 Hz ($t = -15.3$) – and their variability (20–282 Hz σ : $t = +13.3$; 282–639 Hz σ : $t = +12.1$). In D_{ad} the same low-band variability (20–282 Hz σ : $t = +6.8$) is joined by *mid-band* energy means (1.79–2.69 kHz: $t = +6.7$; 1.13–1.79 kHz: $t = +6.7$; 2.69–3.93 kHz: $t = +6.1$), echoing the mid-band shift seen in the association and occlusion profiles. The variance contrast between caught and missed spoofs (TP vs. FN, Levene) is dominated by the low-band energy variability 20–282 Hz σ (Levene $W = 33.6$, $q = 2 \times 10^{-8}$ for D_{zs} ; $W = 72.8$, $q = 7 \times 10^{-16}$ for D_{ad}): spoof clips the detector misses carry markedly different low-band variability from those it catches. These tests significantly link specific frequency bands to the detectors’ error structure, with low-band variability recurring for both detectors.

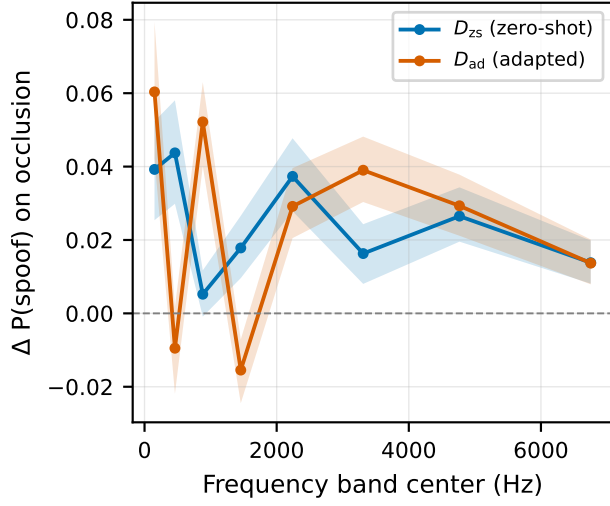


Figure 4: Direct overlay of the two mean occlusion profiles with 95% bootstrap CI ribbons, easing the band-by-band D_{zs} vs. D_{ad} comparison.

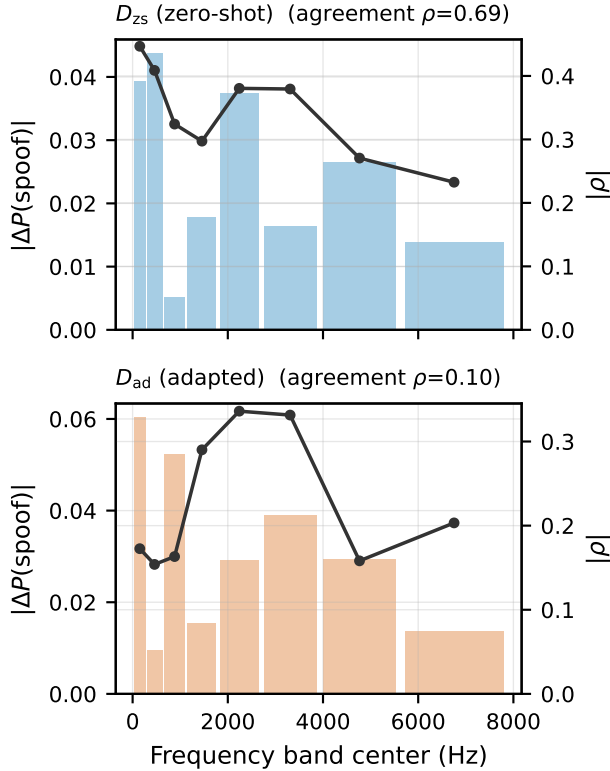


Figure 5: Agreement between the associative and causal analyses on the shared mel-band grid (D_{zs} top, D_{ad} bottom). Bars: magnitude of the causal occlusion drop per band; line (right axis): associative $|\rho|$ per band. Each title reports the band-wise Spearman correlation between the two profiles.

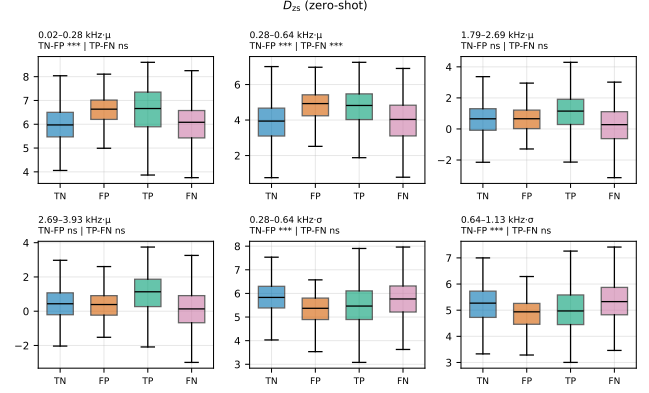


Figure 6: Distribution of the most association-salient (Spearman $|\rho|$) band-energy features across decision quadrants for D_{zs} . Per-panel annotations give the post-FDR significance of the mean contrast (TN vs. FP, Welch) and the variance contrast (TP vs. FN, Levene): *** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$, ns otherwise (H3, inferential). The analogous figure for D_{ad} is confirmatory_box_ad.

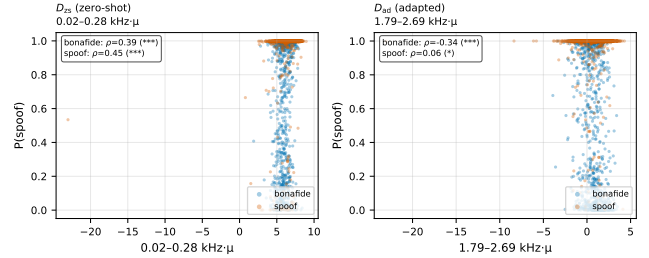


Figure 7: Intra-class association between the most correlated band-energy feature and $P(\text{spoo})$ for each detector. Points are coloured by ground-truth class; Spearman's ρ (with post-FDR significance) is reported per class. Computing ρ within each class avoids the spurious correlation induced by class separation.

As a complementary associative view, Figure 7 relates the most correlated band-energy feature to the detector score within each class (spearman_table). The association is monotonic and class-dependent. For D_{zs} the low-band energy mean (20–282 Hz) rises with $P(\text{spoo})$ in both classes ($\rho = +0.45$ in spoo, $+0.39$ in bonafide; both $q < 10^{-47}$), i.e. higher low-frequency energy co-varies with a higher spoo score. For D_{ad} the dominant cue is the mid-band energy mean (1.79–2.69 kHz): it *lowers* the score among bona-fide clips ($\rho = -0.34$, $q < 10^{-34}$) yet is essentially flat among spoo clips ($\rho = +0.06$), showing the cue acts chiefly by *depressing the score of genuine speech* rather than by discriminating within the spoo class. Computing ρ within each class is therefore essential: pooling the classes would manufacture a spurious global correlation from the score's bona-fide/spoo separation alone.

7 DISCUSSION AND LIMITATIONS

Several caveats bound the interpretation of these results. *Registers do not mix.* The associative analysis and the causal analysis answer different questions; we keep their language separate and treat their agreement as corroboration rather than as a single proof. Only H3 is stated inferentially. *Adaptation is shallow by design.* D_{ad} retrains only a linear read-out on a frozen encoder, so the observed cue shifts reflect a *re-weighting* of pre-existing XLS-R representations, not new features learned for pt-BR; a fully fine-tuned encoder could move cues further, and our conclusions should not be extrapolated to that regime. *Occlusion has a confound.* Band-stop filtering both removes a band's content and injects filtering artefacts; a detector might react to the distortion itself rather than to the missing band, so an occlusion effect localises *where* the model is sensitive without proving the band's content is the sole cause. *Band energy is not phonetics.* Log-mel band energies index where spectral power sits, not phonemes; we use them as acoustic pointers, not as phonetic proof. *The associative analysis is univariate.* Spearman correlations score one band at a time and cannot capture cross-band interactions. Because its D_{ad} profile no longer tracks the causal one (band-wise agreement $\rho = 0.10$), we base band-level claims about D_{ad} on the occlusion analysis. *Power in rare quadrants.* The FP and FN quadrants, though numbering in the hundreds, are the smallest, so small effects there may go undetected. Finally, the study uses a single checkpoint, a single pt-BR corpus, and a fixed ≈ 4 s window; generalisation across models, languages, and durations remains to be established.

8 CONCLUSION AND FUTURE WORK

We presented a population-level, dual-axis acoustic explanation of a black-box XLS-R deepfake detector on Brazilian Portuguese, contrasting a zero-shot model with a lightly adapted one. Associative (intra-class Spearman) and causal (spectral occlusion) evidence, computed on a shared mel-band grid, agree that light adaptation reshapes the detector's acoustic reliance: from a diffuse, broadband use of low-band spectral energy toward a localised low-frequency footprint with an emergent mid-band cue that favours bona fide, alongside a clear EER improvement. Placing both analyses on the same frequency axis let us quantify their agreement directly: it is moderate for the zero-shot detector (band-wise $\rho = 0.69$, though only suggestive at eight bands) and essentially absent after adaptation ($\rho = 0.10$), locating where the two views of the model agree and where they do not. Error-linked acoustic differences are statistically significant under FDR control for both detectors. Future work includes stratifying the analysis by spoof generator (the corpus exposes a generator-method attribute, enabling per-algorithm ANOVA), scaling to the full corpus, correlating band-energy descriptors with the XLS-R embedding space to bridge acoustic and representational views, and extending the protocol to fully fine-tuned detectors and additional languages.

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